

MAKE UP EXAMINATIONS – OCTOBER 2023

Program	: B.E. - Information Science and Engineering	Semester	: IV
Course Name	: Design and Analysis of Algorithms	Max. Marks	: 100
Course Code	: IS43	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Describe the various asymptotic notations with an example for each. CO1 (09)
 - Design an algorithm to search an element in an array using sequential search. Derive the best case, worst case and average case efficiency of this algorithm. CO1 (07)
 - Compare the orders of growth of $n!$ and 2^n . CO1 (04)
- Explain the general plan for analyzing the efficiency of a recursive algorithm. Write the algorithm for solving the tower of Hanoi problem. Derive its efficiency. CO1 (09)
 - Consider the following algorithm: CO2 (06)

ALGORITHM Enigma($A[0..n - 1, 0..n - 1]$)

for $i \leftarrow 0$ to $n - 2$ do

for $j \leftarrow i + 1$ to $n - 1$ do

if $A[i, j] \neq A[j, i]$

return false

return true

- What does this algorithm compute?
 - What is its basic operation?
 - How many times is the basic operation executed?
 - What is the efficiency class of this algorithm?
- Discuss why analysis of the best-case efficiency is not as important as that of worst-case efficiency. CO1 (05)

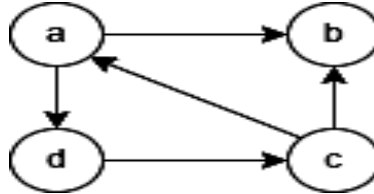
UNIT - II

- Write Bubble sort algorithm and analyze its complexity. CO2 (06)
 - Develop a pseudocode to sort an array using Merge sort technique. Apply the same to sort the list $\{P, A, N, D, E, M, I, C, S, O, N\}$ in alphabetical order. CO2 (10)
 - Analyse the efficiency of binary search algorithm. CO2 (04)
- Define brute force. Apply the sequential search algorithm and Horspool algorithm to determine the number of character comparisons made for searching the pattern "INCREAS" in the text: "THERE_IS_MORE_TO_LIFE_THAN_INCREASING_ITS_SPEED". CO2 (12)

- b) Write an algorithm to sort the elements using quick sort method. Trace quick sort algorithm for the given array of numbers, also show the tree calls and compute average time complexity. 2, 7, 8, 3, 1, 9, 5, 6, 3. CO2 (08)

UNIT - III

5. a) Trace the below graph using Warshall's algorithm. Write the algorithm and indicate why is this an application of dynamic programming. CO3 (08)



- b) What is greedy design technique? Explain the requirements to be satisfied by a greedy choice. CO3 (05)
- c) Write the Kruskal's algorithm for finding the optimal solution for the minimum spanning tree. CO3 (07)
6. a) Apply the dynamic programming algorithm to the following instance of the knapsack problem. The capacity of the knapsack (W) is 5. CO3 (10)

Item	Weight	Value (\$)
1	2	12
2	1	20
3	3	15
4	2	15

Write an algorithm for constructing a table for solving the knapsack problem using dynamic programming.

- b) Indicate the algorithm that can be used to find the shortest paths between a source vertex to all of its other vertices. Write the algorithm for the same. CO3 (10)

UNIT- IV

7. a) Discuss the working of breadth first search approach with an example. CO4 (06)
- b) Develop an algorithm to construct a heap using Bottom-up heap construction. Apply the algorithm to construct a heap for the list {12, 30, 37, 5, 2, 16, 4}. CO4 (07)
- c) Starting at vertex "a" and resolving the ties by vertex alphabetical order, traverse the following graph shown in Fig.7(c) below using depth first search algorithm. Write the DFS forest and show the stack ordering. Discuss its time complexity. CO4 (07)

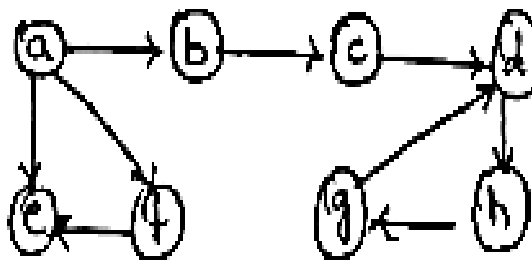


Fig. 7(c)

8. a) What do you understand by Topological sorting? Explain any one method to solve Topological sorting problem with an example. CO4 (06)
- b) Perform heap sort on the list 9, 5, 8, 3, 2, 4, 7, 0, 1, 6 by showing every step of heap construction and heap sort. CO4 (08)
- c) Show the construction of AVL trees for the list 5,6,8,3,2,4,7, by successive insertions. CO4 (06)

UNIT - V

9. a) What do you understand by the term backtracking? Discuss how backtracking is used for solving 4-Queen problem. Show the state space tree for the same. CO5 (12)
- b) Distinguish between backtracking and branch and bound algorithm? CO5 (08)
10. a) Write short notes on P, NP and NP complete problems. CO5 (08)
- b) Consider the Travelling Sales person instance defined by the following cost matrix. CO5 (12)

∞	3	1	5	8
3	∞	6	7	9
1	6	∞	4	2
5	7	4	∞	3
8	9	2	3	∞

Draw the state space tree and show the reduced matrices corresponding to each of the node.
